

The architecture is divided into **three main diagrams**:

1. **System Node Architecture**
 2. **ROS2 Topic Communication Architecture**
 3. **Package & Node Hierarchy**
-

System Architecture

6 layers:

1. User Interaction Layer

- Touchscreen
- Voice Input
- QR Interaction
- Audio Output

2. AI Interaction Layer

- NLP Processor
- Intent Classifier
- Response Generator
- Tour Dialogue

3. Navigation Layer

- Destination Manager
- Waypoint Manager
- Nav2 Planner
- Nav2 Controller
- BT Navigator

4. System Management Layer

- State Manager
- Battery Monitor
- Diagnostics
- Network Monitor
- Tour Manager

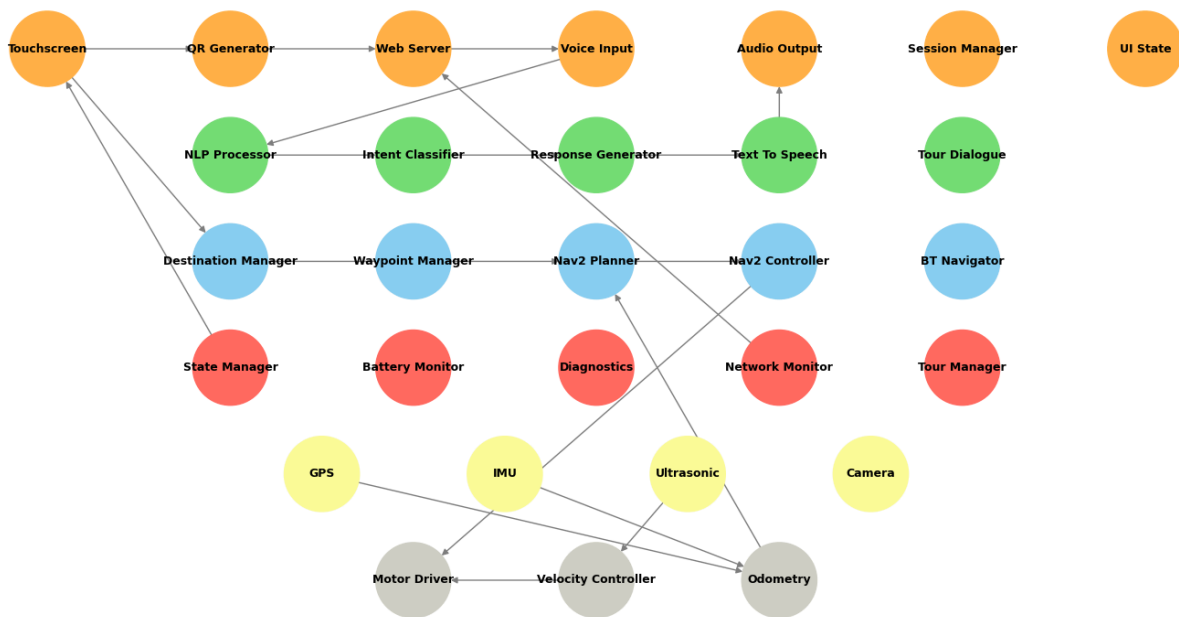
5. Sensor Layer

- GPS
- IMU
- Ultrasonic
- Camera

6. Control Layer

- Velocity Controller
- Motor Driver
- Odometry

Node Architecture Explanation



Color Meaning:

Color	Layer
Orange	User Interface
Green	AI Interaction

Blue	Navigation
Red	System Management
Yellow	Sensors
Grey	Control

User Interface Nodes

Touchscreen

Function

- Primary user interaction
- Destination selection
- Voice trigger
- Tour selection

Publishes

/ui/destination_selected
/ui/mic_button_pressed

Subscribers

/ui/screen_state

QR Generator

Function

- Generates QR for phone voice input
- Creates session IDs

Publishes

/voice/qr_code

Service

/session/create_voice_session

Web Server

Function

- Phone UI
- Voice recording
- Web interface

Publishes

/voice/text_input

Voice Input

Function

- Accepts voice input
- Converts speech to text

Publishes

/voice/text_input

Audio Output

Function

- Speaker output
- Phone audio output

Subscribers

/robot/response_text

Session Manager

Function

- Multi-user handling
 - Voice session tracking
-

UI State

Function

- UI state transitions
 - Screen updates
-

AI Interaction Layer

NLP Processor

Function

- Process text
- Extract meaning

Input

/voice/text_input

Output

/intent

Intent Classifier

Function

- Understand user intent

Examples:

User says:

- "Take me to library"

- "Tell me about IITGN"

Classifier outputs:

NAVIGATION

QUESTION

TOUR

Response Generator

Function

- Generate robot response

Example:

"Sure, taking you to the library"

Text To Speech

Function

- Convert text to speech

Publishes:

Audio stream

Tour Dialogue

Function

- Tour script handling
- Location explanations

Example:

"This is IITGN Academic Block"

Navigation Layer

Destination Manager

Function

- Handles predefined locations
- Converts location name → coordinates

Example:

Library → GPS coordinates

Waypoint Manager

Function

- Multi-stop navigation
 - Tour path planning
-

Nav2 Planner

Function

- Global path planning

Example:

A → B shortest path

Nav2 Controller

Function

- Local trajectory control
 - Velocity generation
-

BT Navigator

Function

- Behavior tree execution
 - Recovery behaviors
-

System Management

State Manager

Function

Robot state machine

States:

- Idle
 - UI Interaction
 - Destination Selected
 - Navigating
 - Speaking
 - Charging
-

Battery Monitor

Function

- Battery percentage
 - Low battery alerts
-

Diagnostics

Function

- System health monitoring
- Error detection

Network Monitor

Function

- WiFi monitoring
 - Internet availability
-

Tour Manager

Function

- Tour route management
 - Tour execution
-

Sensor Layer

GPS

Function

- Global positioning

Topic:

/gps/data

IMU

Function

- Orientation
- Motion detection

Topic: /imu/data

Ultrasonic

Function

- Obstacle detection

Topic:

/obstacle

Camera

Function

- Vision
 - Future object detection
-

Control Layer

Velocity Controller

Function

- PID velocity control
-

Motor Driver

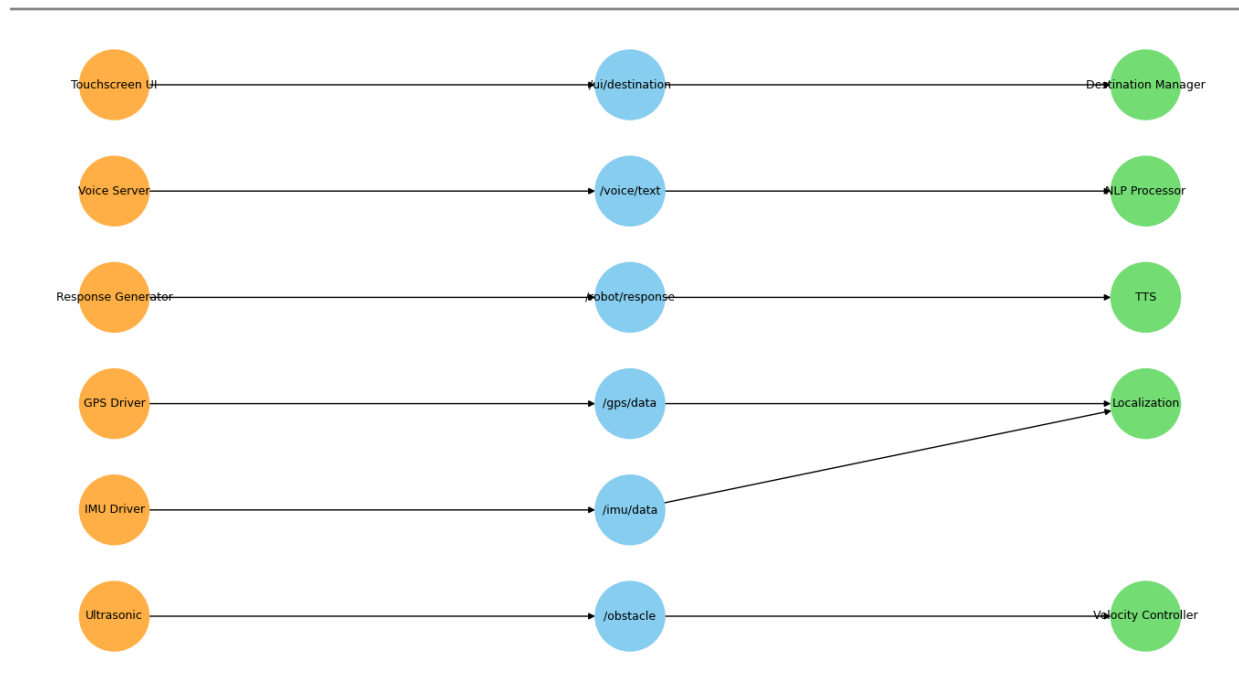
Function

- Motor PWM control
-

Odometry

Function

- Position estimation



Key Topics

UI Topics

/ui/destination
/ui/mic_button
/ui/screen_state

Voice Topics

/voice/text
/voice/qr

Navigation Topics

/gps/data
/imu/data

/obstacle

Control Topics

/cmd_vel

/odometry

Package Description

chitti_system

Core system nodes:

- state_manager
 - diagnostics
 - battery_monitor
-

chitti_hmi

User interface:

- touchscreen
 - QR generator
 - web server
-

chitti_interaction

AI interaction:

- NLP
- intent classifier

- response generator
-

chitti_navigation

Navigation:

- Nav2
 - waypoint manager
 - destination manager
-

chitti_sensors

Hardware drivers:

- GPS
 - IMU
 - Ultrasonic
-

chitti_control

Robot motion:

- Motor driver
 - Velocity controller
 - Odometry
-

chitti_msgs

Custom messages

Example:

LocationRequest
RobotState
TourStatus

chitti_simulation

Gazebo simulation

System Data Flow

User Interaction Flow:

User touches screen
↓
Destination Selected
↓
Destination Manager
↓
Nav2 Planner
↓
Nav2 Controller
↓
Velocity Controller
↓
Motor Driver
↓
Robot Moves

Voice Interaction Flow

User scans QR
↓
Phone voice input
↓
Web Server
↓

NLP Processor
↓
Intent Classifier
↓
Response Generator
↓
Text To Speech
↓
Audio Output

13. Sensor Fusion Flow

GPS
IMU
Ultrasonic
↓
Localization
↓
Nav2

14. Robot State Flow

States:

Idle
UI Interaction
Destination Selected
Navigating
Speaking
Charging